Balanced forces can go with steady motion

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Inertia / ES: inercia / PT: inércia

More mass resists velocity change more.

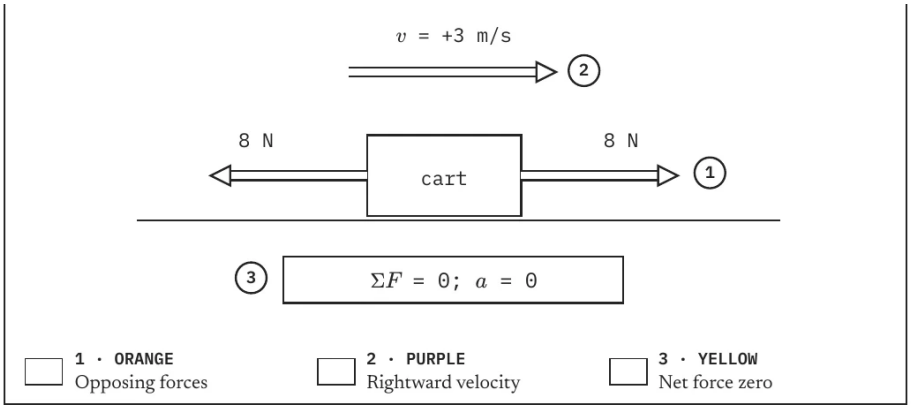
Also on your note page: Inertial frame, Net force, Equilibrium.

B READ THE RELATIONSHIP

Balanced forces → zero acceleration → constant velocity

As you read, find: balanced · steady · changes.

C TRACE THE ORIGINAL FIGURE



Trace equal forces first. Then trace velocity. In pencil, label the arrows F or v.

D FOLLOW THE WORKED EXAMPLE

A cart is at −2 m , moving at +3 m/s . Its external forces balance for the next 4 s . Find its final position.

Given x 0 = −2 m v i = +3 m/s Σ F = 0; t = 4 s

Rule First law: constant v . x f = x 0 + vt .

Work a = 0 m/s² x f = −2 + 3(4)

Answer +10 m Net force zero; motion continues.

E SAVE FOR THE FINAL MEETING

XJ is moving. A forward push is not needed to keep it moving. Gravity can still change its velocity.

Read the physics: Inertia

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Push a cart across a rough floor, then let go. It slows because friction still acts on it. The floor and the cart interact. Inertia is a property of the cart; it does not push.

02 HOW IT WORKS

A force is a push or pull. Net force means all the forces on one object added with their directions. When the forces balance, acceleration is zero. A moving cart keeps its speed and direction. A resting cart stays at rest.

03 WHAT TO CHECK

Use a frame that is not speeding up or turning. A purple velocity arrow shows motion. Orange arrows show forces. Never add the purple arrow to the orange arrows.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart is at −2 m , moving at +3 m/s . Its external forces balance for the next 4 s . Find its final position. +10 m Net force zero; motion continues.

Model check: Balanced forces → zero acceleration → constant velocity.

05 CONNECT TO XJ

XJ needs no engine to keep moving. Over a short interval, use steady velocity as a model. A gravity pull can change the speed or direction. A new observation can test the model.

SAY IT BEFORE YOU WRITE

Is the cart moving, or is its velocity changing? How can you tell? The cart moves ___. Its velocity ___ because the forces ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Inertial frame

ES: sistema de referencia inercial / PT: referencial inercial

Meaning in my words

How would you explain inertial frame to a friend?

In my words, this means ___.

How can you tell?

A puck glides straight at steady speed. Would a camera moving straight at constant speed also see steady motion?

I can tell because ___.

Example or sketch

Which bus ride could serve as an inertial frame for watching a puck: cruising straight or turning?

My example shows ___.

Non-example and why

A bus brakes and a loose puck seems to surge forward inside it. Does that prove a forward force acted on the puck?

I would change ___ because ___.

Net force

ES: fuerza neta / PT: força resultante

Meaning in my words

How would you explain net force to a friend?

In my words, this means ___.

How can you tell?

Two people push one cart in opposite directions. Why do the arrow directions matter?

I can tell because ___.

Example or sketch

One person pushes 6 N right and another 4 N left. What single arrow replaces both?

My example shows ___.

Non-example and why

A student adds those pushes to get 10 N right. Which push did they count the wrong way?

I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Inertia

ES: inercia / PT: inércia

Meaning in my words	How can you tell?
What does inertia mean? Say it the way you'd explain it to a friend. In my own words, inertia means …	An empty cart and a loaded cart both start at rest. We push each with the same net force for one second. Which gains less speed? Why? The ___ cart gains less speed because …
Example or sketch	**Non-example and why**
A cart is already rolling right. The forces on it balance. Where would you draw it one second later? And one second after that? Label your three carts 0 s, 1 s and 2 s.	Someone points to the hand pushing the cart and says, "That push is inertia." What would you tell them? The hand supplies ___. The cart has ___.

Equilibrium

ES: equilibrio / PT: equilíbrio

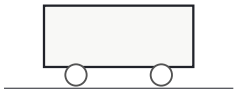
Meaning in my words	How can you tell?
How would you explain equilibrium to a friend? In my words, this means ___.	A cart rolls straight at steady speed. What must the horizontal forces add up to? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch the horizontal forces on a cart with a 3 N pull right and 3 N friction left. My example shows ___.	A tossed ball is briefly still at the top. Is it in equilibrium there? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a moving cart with balanced force arrows in orange and its separate velocity arrow in purple.

Labels you can use: balanced · steady · changes.



Tell a partner

The cart moves ___. Its velocity ___ because the forces ___.

One camera. Which window?

09

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use timed sightings to make a prediction, then check it.

CHOOSE ONE JOB TOGETHER

TRACKING ENGINEER ☐

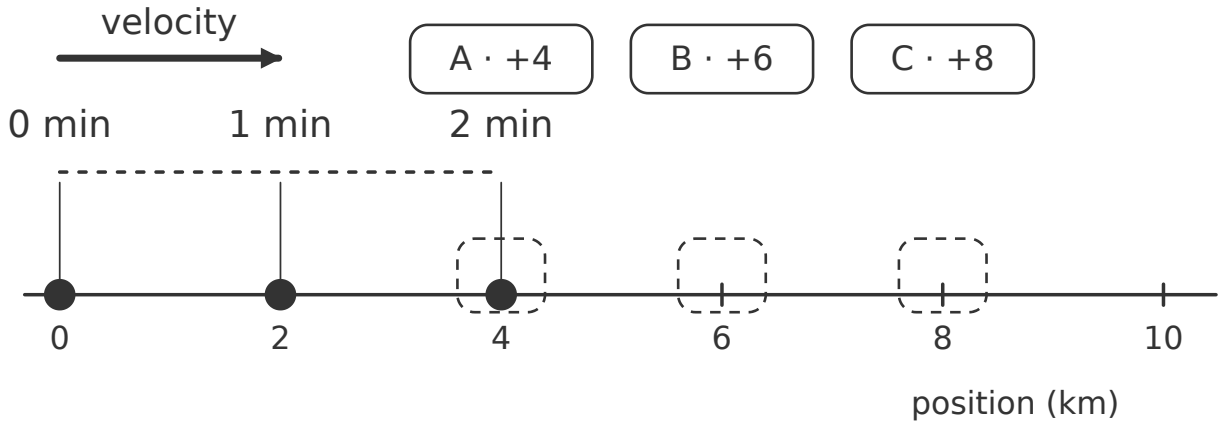
Radio a camera window. Point to the sightings that support your call.

SCIENCE BROADCASTER ☐

Give a **15**-second update. Use your cart drawing to explain why XJ can keep moving without an engine.

Word help: aim / window / coast

XJ is drifting right. At minute **3**, one camera can aim at one window: A, B or C.



01 OUR FIRST CALL

Where should the camera look? Include a signed velocity and say what you expect to stay the same.

Aim at ___ because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

How far did XJ move in the last minute? Use that result in your revised call.

Now we would say ___ because ___.

Switch speakers. Why could a straight-line forecast work for a minute even while gravity pulls on XJ?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks